Varun Ramanathan

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EXPERIENCE

BUSINESS MANAGEMENT SPECIALIST — TD BANK 2025 - Present

- Designed and developed Python automation tools that streamlined enterprise data governance workflows, reducing validation and processing time by over **90%** while improving accuracy and consistency.
- Built SQL-based data validation solutions for enterprise data supply chains hosted within Microsoft Azure cloud environments.
- Developed interactive HTML reports and visualization tools that simplified technical validation results for business and engineering stakeholders.
- Created Power BI dashboards to monitor operational metrics, data quality KPIs, and enterprise reporting.
- Collaborated with cross-functional teams to improve business processes through software automation and technical solutions.

MACHINE LEARNING ENGINEER — QUANTO 2024 - 2025

- Designed and developed backend services in Python powering AI-driven bookkeeping functionality.
- Integrated OpenAI and Google Gemini APIs to deliver intelligent financial analysis and automated bookkeeping capabilities.
- Developed responsive frontend functionality using Next.js and JavaScript, creating seamless user experiences for AI-powered workflows.
- Built real-time data pipelines to process financial data and generate automated insights for small business users.
- Deployed production applications using Vercel and Supabase while collaborating with product and design teams to rapidly deliver customer-facing features

SOFTWARE DEVELOPER — CIBC 2023 - 2024

- Modernized enterprise banking software by migrating legacy C++ and Java applications to Java 21.
- Developed scalable Java backend services and RESTful APIs supporting enterprise cloud migration initiatives.
- Collaborated with architects and stakeholders to design software solutions for global banking applications.
- Built comprehensive unit and integration tests to improve software quality and maintainability.
- Supported cloud deployments, User Acceptance Testing (UAT), production releases, and ongoing application maintenance.
- Diagnosed and resolved production issues while implementing enhancements requested by business stakeholders.

MACHINE LEARNING ENGINEER — MCMASTER 2020 - 2021

- Developed machine learning applications using TensorFlow, OpenCV, and STM X-CUBE-AI for embedded AI and IoT systems.
- Created educational software and demonstration applications supporting machine learning training initiatives.
- Collaborated with academic researchers and industry partners to translate technical concepts into practical software solutions.
- Built Power BI dashboards and visualization tools supporting research analysis.

DATA ANALYST — MCMASTER 2019 - 2020

- Designed and trained TensorFlow computer vision models achieving **95%** vehicle classification accuracy from aerial drone imagery.
- Developed automated image labeling software using Python to reduce manual annotation effort.
- Built machine learning solutions supporting intelligent traffic analysis in partnership with Peel Regional Police.
- Developed embedded network security prototypes focused on anomaly detection and intrusion prevention.

TECHNICAL SKILLS

PROGRAMMING LANGUAGES: Python, Java, JavaScript, SQL, C++, C#, HTML5

AI & MACHINE LEARNING: TensorFlow, OpenAI API, Google Gemini API, Computer Vision, OpenCV, Neural Networks, LLMs, Prompt Engineering, STM X-CUBE-AI

FRONTEND: React, React Native, Next.js, Node.js, AngularJS, HTML, CSS, JavaScript, jQuery

CLOUD SERVICES & PLATFORMS: Azure, AWS, Google Cloud Platform, Supabase, Vercel, Docker, Power BI, Git

OPERATING SYSTEMS: Windows, Linux/Unix, Mac, Android, iOS

TECHNICAL PROJECTS

SIREN SENSE: Designed and developed a cross-platform mobile application using React Native, JavaScript, and Expo that leverages machine learning to detect emergency sirens and hazardous sounds in real time. The application automatically disables active noise cancellation and pauses media playback to improve user awareness and safety. Optimized performance across both iOS and Android devices.

GIGO ROBOTIC ARM: Developed an assistive robotic arm for wheelchair users with Cerebral Palsy, integrating Python, Arduino, Autodesk Inventor, and SolidWorks to enable joystick-controlled object manipulation. Focused on improving accessibility through practical software and hardware integration.

EDUCATION

(M.ENG) **MASTER OF COMPUTER ENGINEERING** — University of Michigan, United States 2021 - 2023
Specialization in Machine Learning & Data Science

(B.ENG) **BACHELOR OF COMPUTER ENGINEERING** — McMaster University, Canada 2017 - 2021